Design and Experimental Evaluation of a Hybrid Tracked Rescue Robot Prototype with Modular Additively Manufactured Mobility Components

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***Abstract*—This paper presents the design and initial evaluation of a 34 kg tracked prototype combining a welded-steel chassis with modular PLA track links, TPU grousers, printed sprockets and four flippers driven by custom three-stage planetary gearboxes of 64:1 reduction. After a printed second-stage output failed during design iteration, comparative linear-static FEA under an estimated $19.2\ \mathrm{N\,m}$ short-duration transmitted stall torque predicted a minimum safety factor below unity for the PLA model and 3.09 for generic aluminum, exceeding the adopted target of 2.0. No failure of the machined replacement was observed during the subsequent operational campaign. The prototype completed 20 trials across ten field-oriented scenarios, accumulating approximately 100 minutes. Four replacements were recorded: three at the gearbox-to-flipper coupling after lateral impacts and one brittle PLA-link fracture under torsional and off-axis loading. All passed their immediate functional checks; active interventions required 10–30 minutes with parts available. Several unquantified stall events occurred in the RV30 main-drive gearmotors; current, voltage, rotational speed, output torque, and track slip were not instrumented, so their cause could not be determined. Complementary full-scale tests yielded peak-load ranges of 6.72 to 7.13 kN in link–grouser compression and 0.48 to 0.64 kN in link tension; these different load paths are not directly comparable. The results prioritize selective reinforcement, lateral and off-axis load resistance and instrumented drivetrain redesign, but do not establish fatigue life, reliability, comparative terrain performance, or deployment readiness.**

***Index Terms*—active flippers, additive manufacturing, experimental evaluation, hybrid structures, material extrusion, tracked rescue robot**

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I. Introduction

Ground robots can reduce human responder exposure in unstable and hazardous search-and-rescue environments [1]. Tracked locomotion distributes load and provides traction on soft and irregular terrain [2], while articulated flippers reconfigure support geometry across uneven terrain. These advantages impose multi-axial loads on links, traction elements, sprockets, transmissions and flipper supports. However, a local failure can immobilize the platform.

Material-extrusion additive manufacturing, known as AM, enables customized and low-volume mobility parts. Yet print orientation and process parameters strongly affect PLA response [3] and a printable file does not establish structural suitability [4]. AM should be used selectively within hybrid load paths and evaluated in the installed geometry.

Prior rescue robots likewise evolved through operational redesign: Quince was adapted for nuclear-plant deployment [5], SHINOBI adopted impact-tolerant sponge crawlers [6], Hector added slip-clutch flipper protection [7], MRL used an in-house planetary drive while iRAP adopted modular flipper and drive units [8], [9], AutonOHM revised shaft–hub connections and transmissions after breakdowns [10], and Athena steel-reinforced printed flipper pulleys after locked-track failures [11]. These reports, however, describe failures and redesigns mostly qualitatively, with limited reporting of full-scale component response, event-specific failure context, or active intervention time.

Accordingly, this work makes three contributions: (i) a 34 kg hybrid tracked architecture combining a welded-steel load path with a separable AM mobility layer whose links can be individually replaced; (ii) exploratory full-scale tests of complete link–grouser assemblies in compression and links in tension using installed geometries and interfaces; and (iii) an exposure campaign relating each replacement event to

TABLE I
PRINCIPAL MECHANICAL SPECIFICATIONS

Parameter	Value
Robot / mobility-system mass	34 / 20 kg
Locomotion	Two main tracks; four flippers
Main drive	2×200 W; 8.35 N·m; 220 rpm
Flipper actuation	$4\times$AX-12A; 64:1 transmission
Overall dimensions	$636 \times 560 \times 274$ mm
Track width / contact length	46 / 470 mm
Ground clearance	36.2 mm

its loading context, functional check, and active intervention time. Selective metal reinforcement of printed drive parts has precedent [11]; the distinction here is reporting architecture, full-scale component response, and operational replacement events together for one prototype. FEA provides complementary design evidence, not validation.

II. PROTOTYPE DESIGN AND FABRICATION

A. Hybrid Mobility Architecture

The prototype uses a welded-steel chassis as its primary load path, as shown in Figure 1. Its 20 kg mobility system comprises two differential main tracks and four independently actuated flippers. Two 200-W RV30 helical-gearmotor assemblies drive the main tracks. Each is rated at a maximum output torque of 8.35 N·m and a no-load output speed of 220 rpm at the gearbox output shaft. A Dynamixel AX-12A actuates each flipper through a custom three-stage planetary transmission that permits continuous flipper rotation. The principal specifications are summarized in Table I.

Figure 2 shows that each main track uses 74 rigid PLA links and each flipper track uses 47, for 336 installed links. A compliant TPU grouser is fastened to each link using M3×10 mm screws. Adjacent links are connected using M4×40 mm screws, and M5×30 mm screws are used in the remaining mechanically fastened assemblies. Any link joint can be opened, so a damaged link can be removed without replacing the complete chain or relying on a single master link. After link replacement, the tensioner is adjusted until the track rotates freely while maintaining full sprocket engagement. Quantitative track-preload measurements were outside the scope of this exploratory prototype evaluation.

Each planetary stage contains a 14-tooth sun gear, four 14-tooth planet gears, and a fixed 42-tooth ring gear, all with module 1.5. With the sun as input and the carrier-side member as output, the nominal reduction per stage is $1 + N_r/N_s = 1 + 42/14 = 4$, producing an ideal three-stage reduction of 64:1, as shown in Figure 3. No lubricant was used.

Most transmission components were printed in PLA as part of the custom AM architecture, enabling in-house fabrication and rapid iteration of the three-stage planetary geometry. A custom transmission was preferred over a commercial gearbox so that its envelope and output interface could match the flipper hub, permit continuous flipper rotation, and allow worn or damaged stages to be reproduced in-house. FFF planetary gearboxes have previously been investigated as a practical approach for rapidly producing custom transmission components, while also recognizing the mechanical limitations of printed polymers [12]. During design iteration, a printed second-stage output failed suddenly under flipper loading before the 20-trial campaign. A subsequent comparative FEA (Sec. III-C) predicted the lowest safety factor around its hexagonal torque-transfer interface. The geometry was retained, and only this component was machined from a laboratory aluminum offcut whose alloy grade and temper were not recorded; one such component was installed per gearbox, while the first- and final-stage outputs remained in PLA. The machined component therefore represented selective reinforcement of the critical torque-transfer interface rather than abandonment of the AM transmission architecture.

B. Material-Extrusion Fabrication

Rigid components were fabricated from 1.75 mm PLA supplied by 4D Lab FLEXCO S.A.S., Colombia and the grousers from the supplier's 1.75 mm TPU. The supplier lists a PLA density of 1.24 g/cm^3, glass-transition temperature near 60 °C, tensile strength of 50–65 MPa and elastic modulus near 3.5 GPa. The listed TPU properties are 95–97 Shore A, 1.30 g/cm^3 density, 16 MPa tensile strength and 330–420% elongation at break [13].

All components were printed on a Bambu Lab P1S with the factory 0.4 mm stainless-steel nozzle using Bambu Studio 2.5.3.61 and the standard P1S, "0.20mm Standard @BBL X1C", and Generic PLA/TPU presets [14]. Layer height was 0.20 mm with four wall loops, five top layers, and three bottom layers. PLA was printed at 220/55 °C and TPU at 240/35 °C (nozzle/bed), with maximum volumetric-flow limits of 12 and 3.2 mm^3/s, respectively. Sparse infill was 30%, except for the corner sprockets (10%). Gyroid infill was used for drive and passive sprockets, flipper shafts, and selected planetary stages; all other components used grid infill. Printing orientation followed the principal load path of each component. Slim tree supports and hair spray were used when required, and no rafts were used. Filament was dried when necessary, although drying conditions were not documented and may affect reproducibility. Visible parts were painted for cosmetic purposes only; test specimens were evaluated unpainted. Table II summarizes the installed AM content.

III. EVALUATION METHODOLOGY

A. Full-Scale Component Tests

Full-scale testing used an Instron universal testing machine. Five complete PLA-link–TPU-grouser assemblies were tested in compression and five complete PLA links in tension. The corresponding crosshead speeds were 1.5 and 4.00 mm/min; each force–displacement curve represents one specimen. For plotting, each force–displacement curve was horizontally translated by subtracting its initial displacement, without scaling the displacement axis or modifying the recorded force values. Table III retains the raw machine-reported displacements, whereas Fig. 6 shows the zero-shifted curves. The

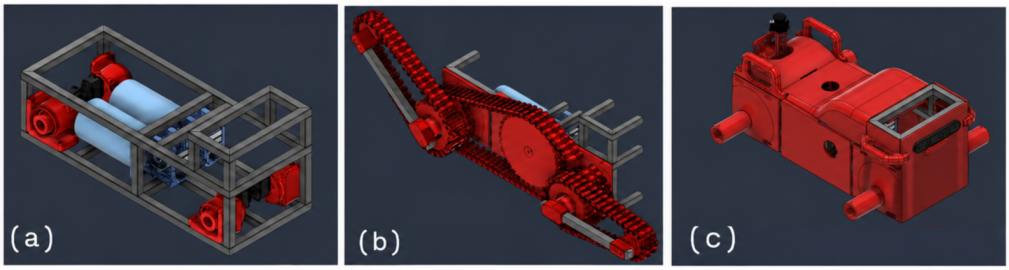


Fig. 1. CAD views of the rescue-robot prototype at three assembly levels: (a) welded-steel chassis and actuation hardware; (b) integrated mobility system with two main tracks and four articulated flippers; and (c) completed 34 kg prototype with bodywork, access panels, protective structures and handling features.

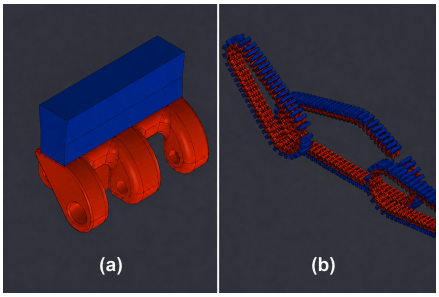


Fig. 2. Modular track construction: (a) one rigid PLA link with its screw-fastened TPU grouser and (b) repeated link–grouser modules connected as an articulated track chain.

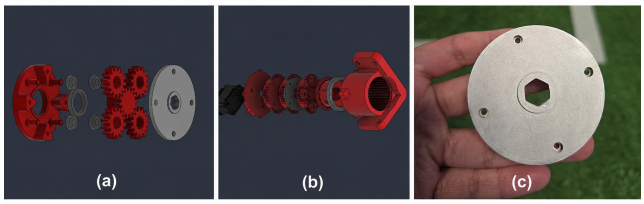


Fig. 3. Three-stage planetary transmission for the flipper mechanism: (a) exploded CAD detail of the second-stage output assembly; (b) exploded CAD view of the complete three-stage planetary drive; and (c) final second-stage output component machined from aluminum.

TABLE II
INSTALLED ADDITIVELY MANUFACTURED CONTENT

Family	Material	Units	Unit mass	Total mass	Print time/unit
Flipper planetary gearbox					
Planet gear	PLA	48	1.94 g	93.12 g	14 min 38 s
Motor-input sun gear	PLA	4	3.52 g	14.08 g	20 min 38 s
Sun gear (stages 2–3)	PLA	8	3.42 g	27.36 g	20 min 7 s
Planet carrier	PLA	12	13.16 g	157.92 g	38 min 5 s
First-stage output	PLA	4	9.40 g	37.60 g	29 min 57 s
Final-stage output	PLA	4	30.16 g	120.64 g	1 h 7 min
Planetary gearbox cover	PLA	4	12.21 g	48.84 g	28 min 47 s
Motor-to-sun coupling	PLA	4	1.40 g	5.60 g	11 min 15 s
Ring gear	PLA	4	120.60 g	482.40 g	4 h 38 min
Track and sprocket assembly					
Front sprocket	PLA	2	231.40 g	462.80 g	7 h 40 min
Rear sprocket	PLA	2	297.63 g	595.26 g	9 h 17 min
Lateral shaft support	PLA	4	184.68 g	738.72 g	4 h 21 min
Flipper drive sprocket	PLA	2	407.90 g	815.80 g	7 h 14 min
Track link	PLA	336	10.68 g	3588.48 g	32 min 41 s
TPU grouser	TPU	336	4.74 g	1592.64 g	16 min 30 s
Drive sprocket spacer	PLA	2	9.57 g	19.14 g	29 min 37 s
Small flipper sprocket	PLA	4	58.00 g	232.00 g	2 h
Mobility and drivetrain AM mass				$\approx 9\,kg$	–
Bodywork and handling features					
Front cover, part 1	PLA	1	96.00 g	96.00 g	2 h 23 min
Front cover, part 2	PLA	1	86.00 g	86.00 g	3 h 5 min
Rear cover	PLA	1	119.00 g	119.00 g	4 h 11 min
Left side cover	PLA	1	129.00 g	129.00 g	3 h 20 min
Right side cover	PLA	1	129.00 g	129.00 g	3 h 20 min
Top cover, part 1	PLA	1	113.00 g	113.00 g	4 h 23 min
Top cover, part 2	PLA	1	87.00 g	87.00 g	3 h 27 min
Top cover, part 3	PLA	1	92.00 g	92.00 g	4 h
Top cover, part 4	PLA	1	84.00 g	84.00 g	3 h 24 min
Front handles	PLA	2	108.50 g	217.00 g	4 h
Bodywork and handling AM mass				$\approx 1.2\,kg$	–
TOTAL INSTALLED AM	—	**791**	—	≈ 10.1 **kg**	≈ 440.8 **h**

Row masses and unit times are slicer estimates. Totals are rounded; time is quantity times unit time and excludes batch-printing effects. Four metallic second-stage outputs are excluded.

retained record lacks load-cell capacity, acquisition frequency, specimen-specific build orientation, fixture-pin specifications and an explicit termination criterion.

Tension was applied through custom metallic fixtures using the installed pin and fastener interfaces; this approximated the nominal tensile path but not the torsional or off-axis operational states. Compression was applied vertically between flat platens to the complete link, attached grouser and screw interface. It therefore characterized the combined assembly rather than TPU alone or a direct service-load condition, as shown in Figure 4.

The specimens used the installed parts' nominal profiles and component-specific settings; no standardized coupons were tested. Results thus characterize the geometries, interfaces and fixtures, not intrinsic material properties. With $n = 5$, results are reported individually and descriptively. The retained records did not isolate compression-failure constituents or tensile-fracture initiation locations.

B. Integrated Operational Trials

Ten field-oriented scenarios were each repeated twice. Each trial comprised approximately 5 min of continuous operation, excluding deployment and initial positioning, for approximately 100 min total. Stair, unstable-gravel and mixed-obstacle cases exercised the tracks and flippers. Representative operational trials are shown in Fig. 5. These exposure trials were not a standardized benchmark; complete dimensions and scenario timings were not retained.

Failures were classified by component and loading context. After the secured, de-energized replacement, track/drive interventions were checked by a 200 mm step traversal and flipper interventions by three lifting cycles. Passing required free

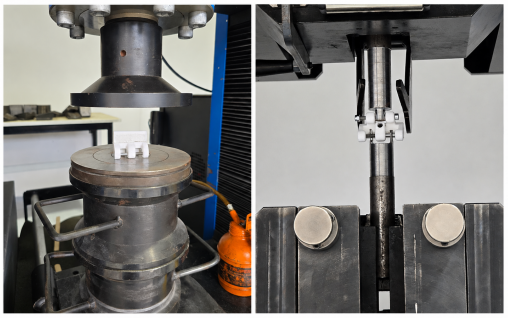


Fig. 4. Full-scale component-test configurations: (a) complete PLA-link–TPU-grouser assembly loaded in compression between flat platens and (b) complete PLA track link loaded in tension through custom fixtures reproducing its pin-and-fastener interfaces.

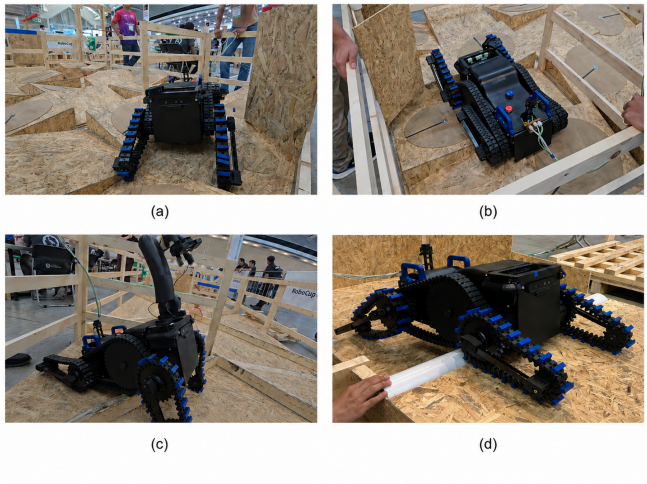


Fig. 5. Representative configurations during the integrated operational campaign: (a) frontal traversal of irregular wooden terrain; (b) confined-course traversal; (c) lateral configuration during obstacle negotiation; and (d) combined main-track and flipper-assisted traversal over mixed obstacles.

rotation, sprocket engagement, secure fasteners and no binding or new visible damage. Active time covered diagnosis through reassembly with the part available, excluding fabrication, tool retrieval and the check. Current, speed, energy and traversal time were not logged consistently.

C. Complementary Finite-Element Assessment

Autodesk Fusion v.2605.1.62 static analyses addressed the second-stage transmission output and a separate principal shaft-support assembly. After the pre-campaign PLA-output failure, the unchanged output geometry was analyzed using a homogeneous isotropic PLA approximation and Fusion's generic-aluminum definition. Both cases used the same constraints and a $19.2\ \mathrm{N\,m}$ remote moment about the transmission axis.

The prescribed moment was calculated as $1.5 \times 16 \times 0.80 = 19.2\ \mathrm{N\,m}$, where $1.5\ \mathrm{N\,m}$ is the nominal AX-12A stall torque, 16:1 is the cumulative reduction of the first

TABLE III
FULL-SCALE COMPONENT LOADING RESULTS

Component	Trial	$F_{\max}$ (kN)	$\delta_{F_{\max}}$ (mm)	δ_{final} (mm)
Complete PLA-link–TPU-grouser assembly, compression	1	7.13	2.00	2.13
	2	6.97	2.07	2.13
	3	6.77	2.07	2.29
	4	6.87	2.31	2.15
	5	6.72	2.25	2.39
Complete PLA track link, tension	1	0.62	1.47	2.56
	2	0.64	3.13	15.12
	3	0.62	3.31	4.68
	4	0.63	3.62	3.99
	5	0.48	1.15	1.85

$\delta_{F_{\max}}$ is displacement at peak load; final displacement is the end of the recorded test. No discrete rupture load was retained.

two stages, and 0.80 is an assumed, unmeasured cumulative mechanical efficiency. It therefore represents an estimated short-duration transmitted stall torque, neither a continuous allowable torque nor a measured service torque. The PLA case used an ultimate-tensile-strength criterion based on first-principal stress, whereas aluminum was evaluated using yield strength and von Mises stress. Parabolic tetrahedral meshes were used, with adaptive refinement configured for up to six steps, a 5% tolerance, and 40% element refinement. The final PLA and aluminum meshes contained 8,295 and 38,919 elements, respectively. The adopted safety-factor target was 2.0.

For the separate support, constrained chassis attachments and a 200 N shaft load represented 2.40 times the static quarter weight under equal four-point sharing. Safety factor was evaluated for both geometries and displacement for the support; mesh, contact and convergence records were not retained. Because the machined alloy is unknown, the aluminum case is a library-material comparison. No causal correspondence is claimed between the 200-N case and the operational events.

IV. Results and Discussion

A. Full-Scale Component Response

Figure 6 and Table III report the individual responses. The five complete PLA-link–TPU-grouser assemblies reached peak compressive loads ranging from 6.72 to 7.13 kN at approximately 2.00 to 2.31 mm of machine-reported displacement. The records ended shortly after the peak; no discrete rupture was confirmed. The load path included the PLA link, TPU grouser, fastened interface and fixture compliance; therefore, the peaks cannot be attributed to TPU alone and the initiating constituent was not isolated. The five PLA links reached peak tensile loads ranging from 0.48 to 0.64 kN at approximately 1.15 to 3.62 mm. Because the two tests used different geometries, interfaces and load paths, their peak loads cannot be used to rank the two configurations as directly comparable component capacities.

B. Integrated Response and Observed Failure Modes

The 20-trial operational campaign, comprising approximately 100 min of operation, is summarized in Table IV. Four

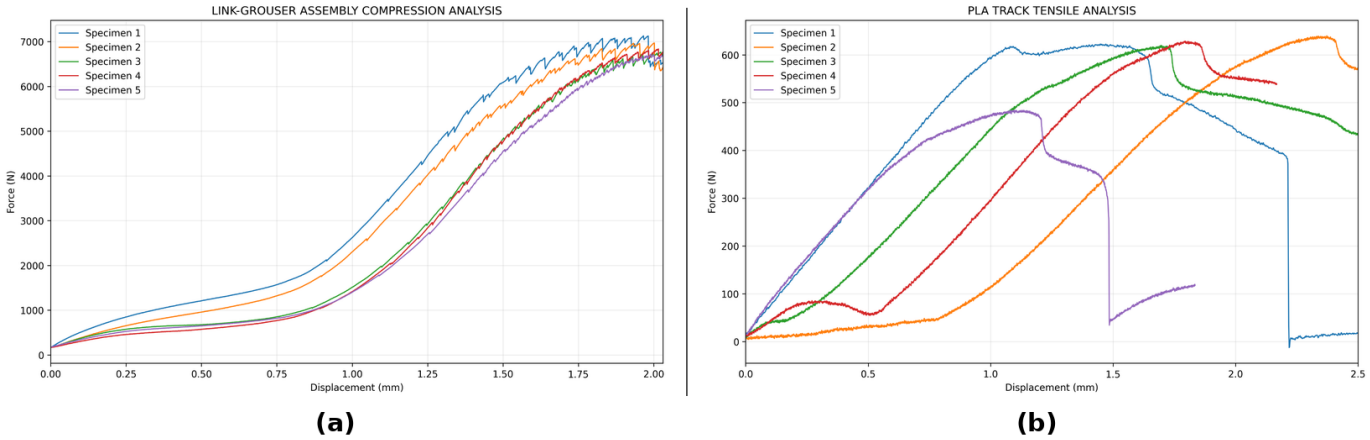


Fig. 6. Force–displacement responses of the full-scale mobility specimens: (a) five complete PLA-link–TPU-grouser assemblies under compression and (b) five complete PLA track links under tension. Each curve corresponds to a different specimen and is shifted to zero initial displacement; panel (b) is truncated at 2.5 mm for display. Table III retains raw values.

component-replacement events were recorded; their event-level record is given in Table IV. Three involved the same gearbox-to-flipper coupling design, which joins a gearbox to its flipper, following lateral impacts. The remaining event was a brittle PLA-link fracture during observed torsional and off-axis loading. All four replacements passed their corresponding immediate functional checks, allowing the exposure trials to resume within the same test session after 10–30 min. No failure of the aluminum replacement for the second-stage output was observed during the campaign.

Several unquantified stall events were observed in the RV30 main-drive gearmotors that actuate the tracks, rather than in the Dynamixel AX-12A flipper servos. Motor current, voltage, rotational speed, output torque, and track slip were not instrumented during these events; therefore, their exact cause could not be determined retrospectively. Although no associated damage or electrical-component failure was observed, the available evidence does not support a definitive diagnosis of the stall mechanism. The campaign also identified three redesign priorities: a stronger lateral load path at the external flipper connection, greater track-link resistance to off-axis loading and an instrumented reassessment of the main drivetrain capacity. The 10–30 min intervention times are descriptive rather than a standardized repairability metric, because access conditions, operator count, alignment and track adjustment varied between events.

TABLE IV
INTEGRATED OPERATIONAL CAMPAIGN AND REPLACEMENT EVENTS

Item	Observed value
Scenarios / repetitions	10 / 2 per scenario
Trials / nominal duration	20 / ≈ 5 min each
Accumulated operation	≈ 100 min
Events 1–3: gearbox-to-flipper coupling	Three replacements after lateral impacts. Damage morphology was not retained; corresponding flipper lifting-cycle checks passed.
Event 4: complete PLA track link	Brittle fracture under torsional/off-axis loading; replacement passed the 200 mm step check.
Post-replacement checks	200 mm step; three lifting cycles
Immediate check outcome	4 of 4 replacements passed
Active intervention time	10–30 min, part available

C. Complementary FEA Results

Figure 7 compares the numerical responses of the second-stage transmission output and the separate shaft-support assembly used at the robot's four principal supports.

For the second-stage output, Fusion reported minimum safety factors of 0.493 and 3.091 for PLA and generic aluminum, respectively [Fig. 7(b),(d)]. The corresponding maximum displacements reported by the simulations were 0.322 and 0.016 mm. The critical response remained concentrated around the hexagonal torque-transfer interface. Thus, PLA failed the adopted criterion, whereas aluminum exceeded the target of 2.0 and reduced predicted displacement by approximately 95%. This contrast qualitatively agrees with the printed-output fracture and subsequent survival of its machined replacement. It supports the material substitution but does not constitute experimental validation of the model, the undocumented alloy, or the unmeasured service torque.

For the separate shaft-support assembly, the 200-N case produced 10.45 MPa maximum von Mises stress, a Fusion-reported minimum safety factor of 4.785 and 0.245 mm maximum displacement [Fig. 7(a),(c)].

D. Design Implications and Limitations

The results support selective reinforcement as a design iteration, not validation of a final reinforced configuration. The internal gearbox-output failure and the external gearbox-to-flipper coupling events involve distinct components and load paths and require separate actions: greater internal torque-transfer capacity and stronger lateral and off-axis resistance at the external connection. Repeated links, grousers, and separable stages remained practical replacement boundaries. A service-load margin for the track links could not be es-

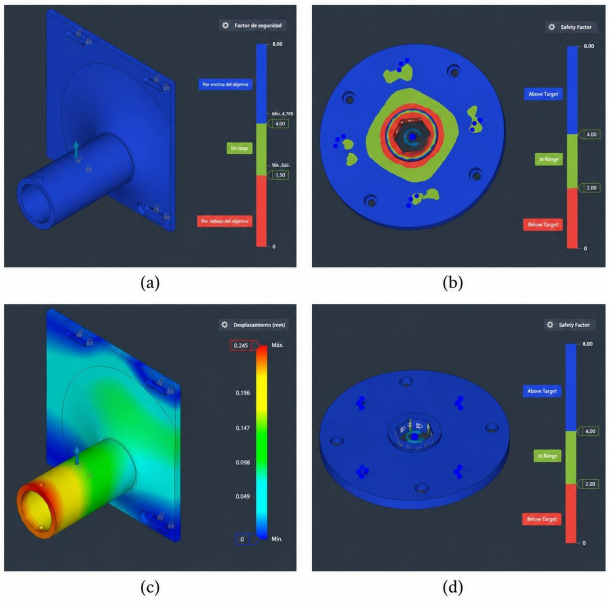


Fig. 7. Complementary FEA: (a) safety factor and (c) displacement of the PLA shaft support under 200 N ($FS_{\min} = 4.785$, $u_{\max} = 0.245$ mm); and second-stage-output safety factor under 19.2 N m for (b) isotropic PLA ($FS_{\min} = 0.493$) and (d) generic aluminum ($FS_{\min} = 3.091$).

tablished because the retained sprocket pitch-radius and load-distribution records were insufficient for a defensible link-force estimate. Overall, the evidence remains prototype-level: it reveals observed failure modes and redesign priorities, not reliability, fatigue life, comparative terrain performance, or deployment readiness. Future evaluation should use standardized rescue-robot scenarios with instrumented drivetrain measurements and physically validate the reinforced transmission under measured service loads using documented, alloy-specific material properties.

V. Conclusions

This work evaluates a 34 kg tracked rescue robot combining a welded-steel load path with modular PLA and TPU mobility components. Full-scale tests yielded 6.72–7.13 kN in link–grouser compression and 0.48–0.64 kN in link tension.

During approximately 100 min of integrated exposure, three external gearbox-to-flipper couplings were replaced after lateral impacts and one PLA link after torsional and off-axis fracture. All four replacements passed their immediate checks; active interventions required 10–30 min. Separately, a preliminary internal output failure motivated aluminum substitution. Under the estimated 19.2 N m transmitted stall torque, Fusion reported minimum safety factors of 0.49 for PLA and 3.09 for generic aluminum; only aluminum exceeded 2.0. The unrefined PLA support model returned 4.79 under 200 N.

Redesign priorities are the internal torque-transfer interface, external flipper load path, links under off-axis loading and instrumented main drivetrain capacity.

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